Weather-Aware Utility Maximization for Urban Attraction Planning

IE 5533 Final Project

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Motivation

- Existing travel platforms usually rank attractions individually.
- Real travelers need a **feasible itinerary**, not only a ranked list.
- Planning must account for:
  - limited time,
  - travel burden,
  - waiting time / congestion,
  - weather uncertainty,
  - budget constraints,
  - overnight stays for multi-day trips.

Main question: How can we generate a high-utility route rather than a simple attraction ranking?

End-to-End System Pipeline

Yelp + Weather + Reviews + Hotels

$\Downarrow$

Feature Engineering

$\Downarrow$

Optimization Model

$\Downarrow$

Route Visualization

- **Predictive layer:** estimate attraction utility, visit duration, waiting time, and cost proxies
- **Prescriptive layer:** optimize tourist routes under time and budget constraints
- **Output layer:** static route summaries and interactive HTML map visualization

Data Sources and Practical Data Gaps

Main data sources

- Yelp business dataset
- Yelp review dataset
- Open-Meteo weather data
- OpenStreetMap accommodation data
- Coordinate-based travel matrices

Important limitations

- incomplete attraction coverage
- no direct queue-length observations
- hotel prices are proxies
- exact road-network travel times are not fully embedded
- some variables rely on open-data approximations

Takeaway: the system is realistic enough for a strong prototype, but not based on a fully complete commercial travel dataset.

Utility and Visit Duration Modeling

Attraction utility

$$U_i = rating_i \cdot \log(review_count_i + 1)$$

- rating measures quality
- review count measures popularity
- log-scaling prevents very popular sites from dominating linearly

Visit duration

$$v_i = \frac{a_i + 4m_i + b_i}{6}$$

- category-based PERT approximation
- different ranges by attraction type
- examples:
  - museums: longer visits
  - parks: medium visits
  - tours / zoos: longest visits

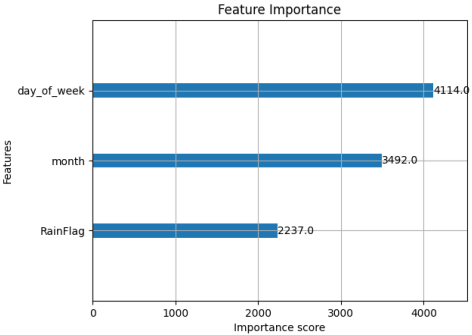
Waiting-Time Prediction

Pipeline

1. Aggregate review density by attraction and date
2. Merge with weather features
3. Train XGBoost to predict review density
4. Convert predicted reviews to visitor counts
5. Compute waiting-time proxy using capacity and attraction type

$$\widehat{V}_{it} = \frac{\widehat{R}_{it}}{0.00078}$$

$$W_{it}^{hist} = b_{\tau(i)} \log\left(1 + \frac{\widehat{V}_{it}}{\kappa_{\tau(i)}}\right) \epsilon_{it}$$



Travel Time and Cost Modeling

Travel-time matrices

- attraction–attraction
- hotel–attraction
- hotel–hotel

$$\tau_{ij} \approx \frac{\text{geodesic distance}_{ij}}{\text{average speed}}$$

Why this matters

- routing feasibility
- travel-time penalty
- multi-day hotel transitions

Cost modeling

- attraction costs are proxy ticket costs
- hotel prices are nightly-price proxies
- both encode relative affordability, not exact market prices

Implementation note

- the attraction cost penalty is treated **nonlinearly**
- in our presentation, we describe it as a squared / stronger marginal penalty for expensive choices

Single-Day Optimization: Objective

Decision variables

$$x_i = \begin{cases} 1 & \text{if attraction } i \text{ is visited} \\ 0 & \text{otherwise} \end{cases} \qquad y_{ij} = \begin{cases} 1 & \text{if travel goes from } i \text{ to } j \\ 0 & \text{otherwise} \end{cases}$$

Objective

$$\max \sum_i U_i x_i - \alpha \sum_i W_i x_i - \beta \sum_{i,j} \tau_{ij} y_{ij} - \gamma \sum_i c_i x_i$$

Interpretation

- maximize attraction utility
- penalize waiting time (congestion)
- penalize travel time between attractions
- penalize expensive itineraries

Single-Day Optimization: Constraints

Resource constraints

$$\text{Time:} \quad \sum_i (v_i + W_i)x_i + \sum_{i,j} \tau_{ij} y_{ij} \leq T$$

$$\text{Budget:} \quad \sum_i c_i x_i \leq B$$

Routing constraints

$$\sum_{j \neq i} y_{ij} = x_i, \qquad \sum_{i \neq j} y_{ij} = x_j$$

Attraction count

$$L \leq \sum_i x_i \leq U$$

MTZ subtour elimination Avoid: A ->B -> C -> A

$$u_i - u_j + U\, y_{ij} \leq U - 1 \qquad \forall i \neq j,\ i,j \neq s$$

$$1 \leq u_i \leq U \qquad \forall i \neq s$$

Profile-Specific Objective Design and Bonus Term

- The route is not optimized with one universal traveler preference.
- We solve separate instances for:
  - **Relaxed**
  - **Balanced**
  - **Explorer**
- Different profiles use different weights and count limits.
- We also include a **bonus term** to shape route behavior across these traveler types.

$$\max \text{ Base Objective} + \lambda \cdot \text{Bonus(selected route/profile)}$$

Role of the bonus term

It is designed to help distinguish the three traveler styles:

- relaxed: lower fatigue, fewer stops
- balanced: moderate trade-off
- explorer: broader attraction coverage

Multi-Day Extension

Real travel planning is usually multi-day, not one single outing.

Multi-day variables

$$x_{i,d}, \qquad y_{i,j,d}$$

Daily time budget

$$\sum_i (v_i + W_i) x_{i,d} + \sum_{i,j} \tau_{ij} y_{i,j,d} \leq T_d$$

Visit-once rule

$$\sum_d x_{i,d} \leq 1$$

Trip budget

$$\sum_{i,d} c_i x_{i,d} \leq B$$

Key idea: attractions must be allocated across days while preserving feasibility and avoiding repeated visits.

Multi-Day Hotel Dataset and Hotel-Aware Routing

Hotel / accommodation layer

- candidates collected from OpenStreetMap
- enriched with Yelp-based review and price-tier proxies
- each hotel receives:
  - nightly-price proxy p_h
  - experience-score proxy e_h

Hotel decision variable

$$z_{h,d} = \begin{cases} 1, & \text{if hotel } h \text{ is selected on day } d \\ 0, & \text{otherwise} \end{cases}$$

Routing interpretation

$$\text{hotel} \rightarrow \text{attractions} \rightarrow \text{hotel}$$

Why it matters

- more realistic for overnight travel
- allows different daily bases
- enables hotel-aware multi-day route design

Full Multi-Day Objective

$$\max \lambda_p^U \sum_d \sum_i U_i x_{id} - \lambda_p^W \sum_d \sum_i W_i x_{id} - \lambda_p^T \left(\sum_d \sum_i \sum_j \tau_{ij} y_{ijd} + \sum_d \sum_h \sum_i a_{hi} s_{hid} + \sum_d \sum_i \sum_h b_{ih} r_{ihd} \right)$$

$$- \lambda_p^C \sum_d \sum_i c_i x_{id} - \lambda_p^H \sum_d \sum_h p_h z_{hd} + \lambda_p^E \sum_d \sum_h e_h z_{hd}$$

- utility contributes positively
- waiting, travel, attraction cost, and hotel cost are penalties
- hotel experience contributes positively
- different profile weights generate different itinerary styles

Multi-Day Optimization: Key Changes

Modified constraints (from single-day)

$$\text{Time (per day):} \quad \sum_i (v_i + W_i) x_{id} + \sum_{i,j} \tau_{ij} y_{ijd} + \sum_{h,i} a_{hi} s_{hid} + \sum_{i,h} b_{ih} r_{ihd} \leq T_d$$

$$\text{Flow (per day):} \quad \sum_{j \neq i} y_{ijd} = x_{id}, \qquad \sum_{i \neq j} y_{ijd} = x_{jd}$$

New constraints

$$\text{Visit once:} \quad \sum_d x_{id} \leq 1$$

$$\text{Hotel start:} \quad \sum_{h,i} s_{hid} = 1$$

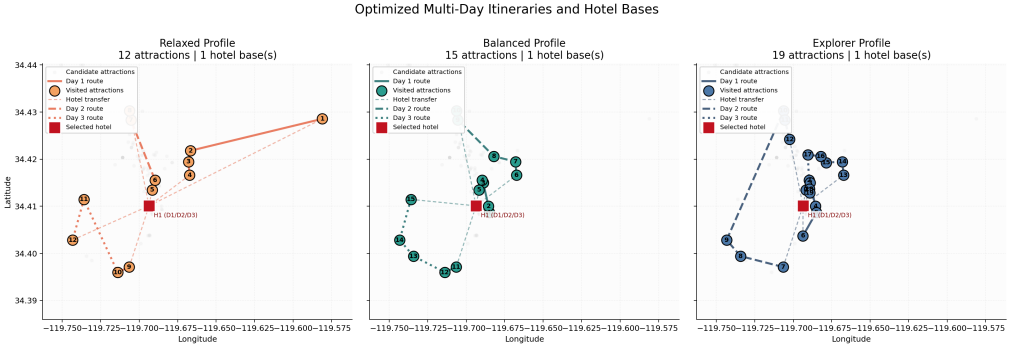
$$\text{Hotel end:} \quad \sum_{i,h} r_{ihd} = 1$$

$$\sum_i s_{hid} = z_{hd}$$

$$\sum_i r_{ihd} = z_{hd}$$

$$\text{Budget:} \quad \sum_d \sum_i c_i x_{id} + \sum_d \sum_h p_h z_{hd} \leq B$$

Static Route Results



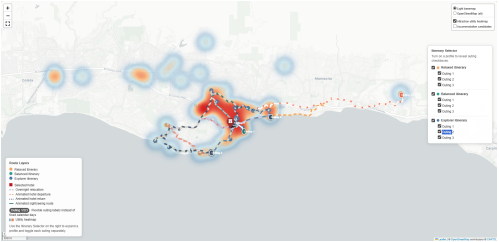
Interpretation

- relaxed routes visit fewer attractions
- balanced routes trade off fatigue and coverage
- explorer routes cover more attractions at higher travel burden

Interactive Route Visualization

Interactive HTML Output

- Our pipeline exports an interactive **HTML map** using Folium.
- Users can explore routes by:
  - traveler profile (Relaxed / Balanced / Explorer)
  - daily itinerary
  - attraction sequence
  - hotel locations
- This visualization helps verify that optimized routes are geographically coherent and practically feasible.



[Open Interactive Map]

Limitations and Practical Challenges

- the dataset is incomplete, so several quantities are proxies
- travel time inside the optimizer is still approximate
- queue length is not directly observed
- hotel prices and attraction costs are not exact market values
- the multi-day mixed-integer model grows quickly with problem size
- for larger instances, the solver may return a feasible but suboptimal solution within the time limit

Takeaway: the system is a strong end-to-end prototype, but there is room for better realism and scalability.

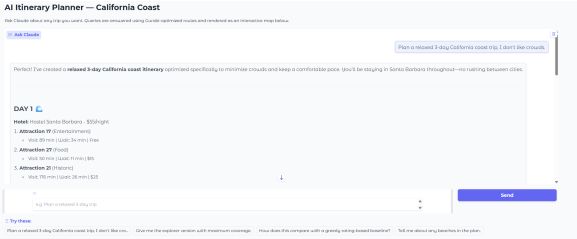
Agent-Based Extension

Why an Extension

- Core contribution: predictive + MIP pipeline
- Optimizer alone isn't user friendly
- Agent layer as a usability extension

What We Built

- Maps natural language to optimizer parameters
- Calls MIP as a backend tool, core unchanged
- Returns itinerary and map link



Future Work and Solution

- We will add the Multi location travel planner.
- To deal with the slow speed, we can design our own solver integrated with benders decomposition and branch and price.
- Integrate some contextual bandits algorithm to deal with the **real time** decision.

Conclusion

- We built an **open-data tourism planning pipeline**.
- We integrated **predictive modeling** and **optimization**.
- We started from a clean **single-day formulation** and extended it to a **multi-day hotel-aware itinerary model**.
- The final system produces both static route summaries and interactive HTML visualization.

Click here: Github link(code and results)

Thank you!